

Optional lecture C

Nesting and the Hubbard model

Objectives. Off the path: nothing on it uses this lecture. Give the particles of a gas a spin, and carrying one round the ring acts on a vector of labels instead of multiplying by a number. The one idea of this lecture is that this eigenvalue problem is itself an integrable model — an inhomogeneous Heisenberg chain whose sites are the gas’s own particles — solved by [Lecture 15](#)’s algebraic Bethe ansatz, with a second set of rapidities. Then the Hubbard model, whose equations have the same shape with k replaced by $\sin k$.

C.1 The question

[Lecture 13](#) gave labelled particles a matrix S-matrix and the Yang–Baxter equation. So the question is

how is a gas of particles with a spin quantised on a ring, when the two-body S-matrix is a matrix?

The answer: the ring condition is an eigenvalue problem for an inhomogeneous Heisenberg chain whose sites are the particles ([Theorem C.2](#) and [Proposition C.4](#)), and its algebraic Bethe ansatz gives two coupled families of Bethe equations ([Theorem C.6](#)). The worked example is two particles ([Example C.7](#)).

C.2 Fermions with a spin, and a matrix S-matrix

The model is [Lecture 11](#)’s delta gas with one addition: each particle carries a spin- $\frac{1}{2}$ label, and the particles are fermions. On a ring of circumference L ,

$$H = - \sum_{j=1}^N \frac{\partial^2}{\partial x_j^2} + 2c \sum_{i < j} \delta(x_i - x_j), \quad c > 0, \quad (\text{C.1})$$

acting on wavefunctions $\psi(x_1\sigma_1, \dots, x_N\sigma_N)$, $\sigma_j \in \{\uparrow, \downarrow\}$, that are *totally antisymmetric* under exchanging a position and its label together. This is the *Yang–Gaudin* model (Yang 1967, Gaudin 1967). Write M for the number of down spins, so $S^z = \frac{N}{2} - M$, and take $M \leq N/2$.

Remark C.1 (Units, and two names used twice). Units are [Lecture 11](#)’s, $\hbar = 2m = 1$. The Hubbard hopping t is a number, the transfer matrix $t(\lambda)$ always carries its argument; the Hubbard Λ_α is a spin rapidity, [Lecture 15](#)’s eigenvalue $\Lambda(\lambda)$ always carries its argument.

The two-body problem is two problems

Lecture 13 did the two-body problem for labelled particles. On $x_1 < x_2$ write

$$\psi(x_1, x_2) = A e^{i(k_1 x_1 + k_2 x_2)} + A' e^{i(k_2 x_1 + k_1 x_2)}, \quad A, A' \in \mathbb{C}^2 \otimes \mathbb{C}^2, \quad (\text{C.2})$$

with slot j of the tensor holding the label of the particle at position j , and extend by antisymmetry. Proposition 13.1 gives $A' = Y_F(k_1 - k_2)A$ with

$$Y_F(k) = -\frac{kP + ic}{k + ic} = S(k) \Pi_- - \Pi_+, \quad S(k) = \frac{k - ic}{k + ic}, \quad (\text{C.3})$$

P the permutation of the two label factors (Definition 9.4) and $\Pi_{\pm} = \frac{1}{2}(1 \pm P)$ the projectors on labels symmetric and antisymmetric — checked.

(C.3) is the whole reason this lecture exists. On the *singlet* — labels antisymmetric, positions symmetric — the delta acts and the amplitude picks up Lecture 11's scalar $S(k)$. On the *triplet* the wavefunction vanishes where the particles meet, the delta never sees them, and the amplitude picks up -1 (Figure C.1). One two-body problem, two answers; a scalar Bethe equation cannot hold both.

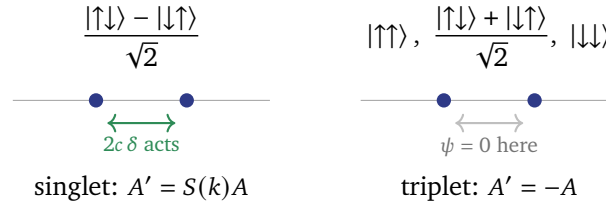


Figure C.1: The two channels of (C.3) are *spin states of the colliding pair*: antisymmetric labels force symmetric positions and the delta bites (left); symmetric labels give a wavefunction vanishing where the particles meet (right). $|\uparrow\downarrow\rangle = (|S\rangle + |T_0\rangle)/\sqrt{2}$ feels both.

Following the momenta

Lecture 13 also fixed the object to work with. Y is written in the order of *positions*; the natural bookkeeping for a scattering problem follows the *momenta*, and there the two-body matrix is $X = PY$ (Equation (13.3)), for fermions

$$X(u) = -\frac{u + iP}{u + i} = \frac{R(-u)}{u + i} = -\frac{L(u)}{u + i}, \quad u = \frac{k_1 - k_2}{c}, \quad (\text{C.4})$$

with $R(u) = u - iP$ the course's rational R-matrix and $L(u) = u + iP$ the Heisenberg Lax operator of Lecture 14. We drop the subscript F: in this lecture X always means the fermionic one. Three facts, all checked symbolically, carry the lecture:

$$X(0) = -P, \quad X(u)X(-u) = 1, \quad X_{12}(u-v)X_{13}(u)X_{23}(v) = X_{23}(v)X_{13}(u)X_{12}(u-v). \quad (\text{C.5})$$

The last is the Yang–Baxter equation (§ 13.3); the first is the only place a sign will come from in § C.3. Note that (C.4) uses $R(-u)$, not R : in the course's RTT ordering (Theorem 14.2) the Lax operator paired with $R(\lambda) = \lambda - iP$ is $\lambda + iP$.

C.3 Periodicity is an eigenvalue problem

Inside the region $x_1 < x_2 < \dots < x_N$ the Bethe ansatz reads

$$\psi|_{x_1 < \dots < x_N} = \sum_{R \in S_N} A_R e^{i \sum_j k_{R(j)} x_j}, \quad A_R \in (\mathbb{C}^2)^{\otimes N}, \quad (\text{C.6})$$

slot j of A_R holding the label of the particle at position j ; total antisymmetry then determines ψ everywhere. The contact conditions at $x_j = x_{j+1}$ are [Lecture 13](#)'s walls, $A_{Rs_j} = Y_{j,j+1}(k_{R(j)} - k_{R(j+1)})A_R$ ([Equation \(13.5\)](#)), and in the momentum-ordered amplitudes $B_R = \Pi_R A_R$ of [Lemma 13.2](#) they become

$$B_{Rs_j} = X_{ab}(u_{ab})B_R, \quad a = R(j), \quad b = R(j+1), \quad u_{ab} = \frac{k_a - k_b}{c}. \quad (\text{C.7})$$

Each crossing of momentum lines a and b multiplies by X_{ab} ([Figure C.2](#)). Nothing is left to choose: the interaction has been used up. What remains is the ring.

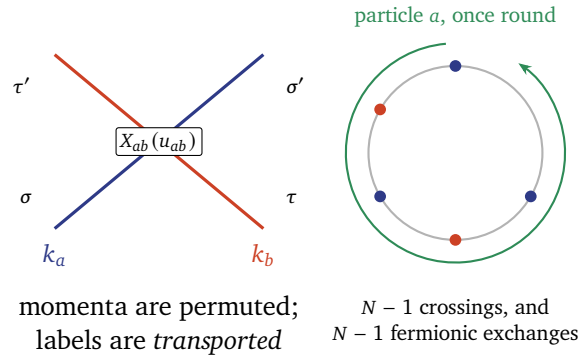


Figure C.2: Left: a crossing is the matrix X_{ab} on the two labels — the momenta are permuted, as classically ([Lecture 12](#)), the labels carried across by an operator. Right: particle a once round the ring crosses every other once and moves past $N-1$ others; both counts appear in [Theorem C.2](#).

Theorem C.2 (Periodicity). Let $k_1, \dots, k_N$ be pairwise distinct, no two differing by $\pm ic$ (so that every $X_{ab}(u_{ab})$ is defined), and let ψ be the totally antisymmetric extension of (C.6) with amplitudes satisfying (C.7). Then ψ is periodic in every variable if and only if, for every $R \in S_N$ and $a = R(1)$,

$$(-1)^{N-1} e^{ik_a L} X_{a,R(N)}(u_{a,R(N)}) \cdots X_{a,R(2)}(u_{a,R(2)}) B_R = B_R. \quad (\text{C.8})$$

These $N!$ conditions are equivalent to N conditions on a single vector $B = B_{\text{id}}$ · sketched here; proved in the reading below,

$$(-1)^{N-1} e^{ik_a L} \mathcal{X}_a B = B, \quad \mathcal{X}_a = X_{a,a-1} \cdots X_{a,1} X_{a,N} \cdots X_{a,a+1}, \quad (\text{C.9})$$

every factor at its own argument $u_{ab} = (k_a - k_b)/c$.

Proof. Because ψ is antisymmetric, periodicity in one argument at every configuration gives it in all of them; take the first. Let $0 \leq x_1 < \dots < x_N < L$ and shift x_1 by L . The shifted point has its arguments in the order $x_2, \dots, x_N, x_1 + L$, so moving the first argument past the other $N-1$ costs $(-1)^{N-1}$:

$$\psi(x_1 + L, \sigma_1; \dots) = (-1)^{N-1} \psi(x_2 \sigma_2, \dots, x_N \sigma_N, x_1 + L, \sigma_1),$$

and both sides of $\psi(x_1+L, \dots) = \psi(x_1, \dots)$ are now values of (C.6). Matching the coefficient of each plane wave: the term R on the right contributes $e^{ik_R(N)L}$ and the momentum word $(R(N), R(1), \dots, R(N-1))$, while the slots of A_R are read in the cyclic order $(\sigma_2, \dots, \sigma_N, \sigma_1)$. Writing c for the cycle $1 \mapsto N, j \mapsto j-1$, and $C = \Pi_{c^{-1}}$ for the operator that permutes slots that way, the condition is

$$A_R = (-1)^{N-1} e^{ik_a L} C A_{Rc^{-1}}, \quad a = R(1). \quad (\text{C.10})$$

Now pass to $B_R = \Pi_R A_R$. Since Π is a homomorphism, $\Pi_{Rc^{-1}}^{-1} = \Pi_c \Pi_R^{-1} = C^{-1} \Pi_R^{-1}$, so $C A_{Rc^{-1}} = \Pi_R^{-1} B_{Rc^{-1}}$ and the relabelling operator cancels: (C.10) is $B_R = (-1)^{N-1} e^{ik_a L} B_{Rc^{-1}}$. The permutation Rc^{-1} is R with its momentum word shifted one place left, reached from R by $s_1 s_2 \dots s_{N-1}$, and at each step (C.7) applies with first leg a throughout: $B_{Rc^{-1}} = X_{a,R(N)} \dots X_{a,R(2)} B_R$. That is (C.8).

The conditions for different R reduce to (C.9): those with the same a are conjugate by the Yang–Baxter equation, and a chain of the X 's carries the others onto B_{id} (below). ■

Proof of the reduction, and the two places it can go wrong (★). Same a. If $R' = R s_j$ with $j \geq 2$, so that R and R' share $a = R(1)$ and differ by exchanging two letters b, b' of the word, then $B_{R'} = X_{bb'} B_R$, and the Yang–Baxter equation (C.5) in the form

$$X_{ab} X_{ab'} = X_{bb'} X_{ab'} X_{ab} X_{bb'}^{-1} \quad (\text{C.11})$$

conjugates the whole product: the operator in (C.8) for R' is $X_{bb'}$ times the one for R times $X_{bb'}^{-1}$, because the remaining factors $X_{a,l}$ with $l \neq b, b'$ have legs disjoint from b, b' . So the two conditions are the same condition. **The order in (C.11) matters:** with the two $X_{bb'}$ exchanged the identity is false, and at $N = 3$ with $(\xi_a, \xi_b, \xi_{b'}) = (\frac{1}{3}, -\frac{5}{7}, \frac{2}{5})$ the two sides differ by an exact rational 0.43 in the largest entry · *checked*.

Different a. Every R with $R(1) = a$ is therefore equivalent to the cyclic word $C_a = (a, a + 1, \dots, N, 1, \dots, a - 1)$, with $C_1 = \text{id}$; but that is a condition on B_{C_a} , not on B_{id} . They are the same condition, by an induction on a . The word C_{a+1} is C_a with its first letter moved to the back, reached by $s_1 \dots s_{N-1}$, and the crossings accumulated on the way are exactly $X_{a,C_a(N)} \dots X_{a,C_a(2)} = X_a$, so

$$B_{C_{a+1}} = X_a B_{C_a}. \quad (\text{C.12})$$

If the condition for a holds, X_a acts on B_{C_a} as the non-zero scalar $q_a = (-1)^{N-1} e^{-ik_a L}$, so $B_{C_{a+1}}$ is a non-zero multiple of B_{C_a} . Starting from $C_1 = \text{id}$, every B_{C_a} is a multiple of B and the N conditions are (C.9); conversely (C.12) runs the same induction backwards. All $N!$ conditions, and the chain, are verified at $N = 3$ and $N = 4$ · *checked*. ■

The fermionic sign in (C.9) is not cosmetic: at $N = 2$ dropping it changes $e^{ikL} = (u+i)/(u-i)$ to $e^{ikL} = -(u+i)/(u-i)$, and moves the root by a factor 2.007 at $L = 10, c = 3$ · *checked*.

The product of crossings is a transfer matrix

(C.9) is N eigenvalue problems for N different operators, which looks hopeless. It is not: the N operators are values of one commuting family, and it is a family we have already solved.

Definition C.3 (The auxiliary chain). For momenta as in Theorem C.2, put $\xi_j = k_j/c$, adjoin an auxiliary $\mathbb{C}^2$ labelled a , and set

$$T_a(\lambda) = X_{aN}(\lambda - \xi_N) \dots X_{a1}(\lambda - \xi_1), \quad t(\lambda) = \text{tr}_a T_a(\lambda). \quad (\text{C.13})$$

Its quantum spaces are the *labels of the particles*, its N “sites” the particles themselves, at inhomogeneities equal to their own momenta: no lattice, no spin Hamiltonian, a bookkeeping device that happens to be integrable.

Proposition C.4 (The crossings are the transfer matrix). *For every j , $t(\xi_j) = -X_j$.*

Proof. At $\lambda = \xi_j$ the j th factor of (C.13) is $X_{aj}(0) = -P_{aj}$:

$$T_a(\xi_j) = -X_{aN} \cdots X_{a,j+1} P_{aj} X_{a,j-1} \cdots X_{a1}.$$

Push P_{aj} to the right using $P_{aj}X_{al} = X_{jl}P_{aj}$ for $l \neq j$; the factors to the right of it become operators on the quantum spaces alone, $\mathcal{M} = X_{j,j-1} \cdots X_{j1}$, which involves only the spaces $1, \dots, j$ and so commutes with $X_{aN} \cdots X_{a,j+1}$. Hence $T_a(\xi_j) = -\mathcal{M} X_{aN} \cdots X_{a,j+1} P_{aj}$, and $\text{tr}_a(F_a P_{aj}) = F_j$, valid for any F not acting on space j , gives $t(\xi_j) = -\mathcal{M} X_{jN} \cdots X_{j,j+1} = -X_j$. ■

Proposition C.5 (The chain is Heisenberg). $T_a(\lambda) = (-1)^N \left[\prod_n (\lambda - \xi_n + i) \right]^{-1} L_{aN}(\lambda - \xi_N) \cdots L_{a1}(\lambda - \xi_1)$: the auxiliary chain is an inhomogeneous Heisenberg chain, up to one scalar. Consequently $[t(\lambda), t(\mu)] = 0$ and $[t(\lambda), S^z] = 0$ for all λ, μ away from the poles $\xi_n - i$.

Proof. The displayed identity is (C.4) factor by factor. The RTT relation (Theorem 14.2) needs only that every site operator obeys RLL with one common R , and the shifts cancel in the argument of R , $(\lambda - \xi_n) - (\mu - \xi_n) = \lambda - \mu$, so commutativity follows as in Theorem 14.3; scalars commute with everything · checked. ■

So all N conditions can be imposed at once: ask for a common eigenvector of the family $t(\lambda)$ and read off the N numbers $t(\xi_a)$. *This is the whole of nesting.* The first Bethe ansatz reduced the gas to a finite linear-algebra problem in the labels; the second solves that problem.

C.4 The nested Bethe equations

Lecture 15 solved this chain for equal inhomogeneities, and nothing in the solution used that they are equal: with $L_{an}(\lambda - \xi_n)$ at site n the RTT relation holds as before (Theorem 14.2 needs only a common R), the exchange relations are unchanged, and the reference-state eigenvalues (15.4) become $\prod_n (\lambda - \xi_n + i)$ and $\prod_n (\lambda - \xi_n)$ · checked, $N = 4$. Work with the unnormalised monodromy of Proposition C.5,

$$\tilde{T}(\lambda) = L_{aN}(\lambda - \xi_N) \cdots L_{a1}(\lambda - \xi_1) = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad (\text{C.14})$$

whose entries are the A, B, C, D below — *not* those of (C.13), which differ from them by the scalar and would give $A|0\rangle = (-1)^N |0\rangle$. Its reference state is all labels up, $|0\rangle$, on which $\tilde{T}$ is triangular with $A(\lambda)|0\rangle = \prod_n (\lambda - \xi_n + i)|0\rangle$ and $D(\lambda)|0\rangle = \prod_n (\lambda - \xi_n)|0\rangle$; B creates; and for pairwise distinct $v_1, \dots, v_M$, none equal to 0 or $-i$, the vector $\Psi = B(v_1) \cdots B(v_M)|0\rangle$ is an eigenvector of the whole family, with

$$\Lambda(\lambda) = \prod_n (\lambda - \xi_n + i) \prod_\alpha \frac{\lambda - v_\alpha - i}{\lambda - v_\alpha} + \prod_n (\lambda - \xi_n) \prod_\alpha \frac{\lambda - v_\alpha + i}{\lambda - v_\alpha}, \quad (\text{C.15})$$

provided the residues at $\lambda = v_\alpha$ cancel (Corollary 15.9).

The spin rapidities below are $\lambda_\alpha = \nu_\alpha + \frac{i}{2}$, minus [Lecture 10](#)'s convention; for generic inhomogeneities the mirrored roots do not solve the spin equations, and the reading at the end of this lecture says why the sign is not free.

Theorem C.6 (The nested Bethe equations, Yang and Gaudin 1967 · *proved, given the reduction of Theorem C.2, sketched here*). Let $k_1, \dots, k_N$ be pairwise distinct with no two differing by $\pm ic$ (so that no X in (C.9) is singular), let $\lambda_1, \dots, \lambda_M$ be pairwise distinct and $\neq \pm \frac{i}{2}$ ([Corollary 15.9](#)'s regular roots, $\nu_\alpha \neq 0, -i$), put $\xi_j = k_j/c$, and let

$$e^{ik_j L} = \prod_{\alpha=1}^M \frac{\xi_j - \lambda_\alpha + \frac{i}{2}}{\xi_j - \lambda_\alpha - \frac{i}{2}}, \quad j = 1, \dots, N, \quad (\text{C.16})$$

$$\prod_{j=1}^N \frac{\lambda_\alpha - \xi_j + \frac{i}{2}}{\lambda_\alpha - \xi_j - \frac{i}{2}} = \prod_{\beta \neq \alpha} \frac{\lambda_\alpha - \lambda_\beta + i}{\lambda_\alpha - \lambda_\beta - i}, \quad \alpha = 1, \dots, M. \quad (\text{C.17})$$

Then $\Psi = B(\lambda_1 - \frac{i}{2}) \cdots B(\lambda_M - \frac{i}{2}) |0\rangle$, if it is not zero, satisfies every periodicity condition (C.9), and the resulting eigenfunction of (C.1) has

$$E = \sum_{j=1}^N k_j^2, \quad K = \sum_{j=1}^N k_j, \quad S^z = \frac{N}{2} - M. \quad (\text{C.18})$$

Proof. (C.17) is [Corollary 15.9](#)'s system (15.17) for the inhomogeneous chain, written in $\lambda_\alpha = \nu_\alpha + \frac{i}{2}$: the residue of (C.15) at ν_α vanishes exactly when $\prod_n (\nu_\alpha - \xi_n + i) / \prod_n (\nu_\alpha - \xi_n) = \prod_{\beta \neq \alpha} (\nu_\alpha - \nu_\beta + i) / (\nu_\alpha - \nu_\beta - i)$. So Ψ is a common eigenvector and [Proposition C.4](#) turns (C.9) into N scalar equations, $(-1)^{N-1} e^{ik_\alpha L} \cdot (-\Lambda^X(\xi_\alpha)) = 1$, with Λ^X the eigenvalue of (C.13). Evaluate it. By [Proposition C.5](#), $\Lambda^X(\lambda) = (-1)^N \Lambda(\lambda) / \prod_n (\lambda - \xi_n + i)$; at $\lambda = \xi_\alpha$ the second term of (C.15) carries the factor $\prod_n (\xi_\alpha - \xi_n) = 0$ and dies, and the first term's prefactor cancels the denominator *exactly*, leaving

$$\Lambda^X(\xi_\alpha) = (-1)^N \prod_{\alpha} \frac{\xi_\alpha - \lambda_\alpha - \frac{i}{2}}{\xi_\alpha - \lambda_\alpha + \frac{i}{2}}. \quad (\text{C.19})$$

Then $(-1)^{N-1} \cdot (-1) \cdot (-1)^N = 1$ for every N , and what is left is (C.16). Inside the sectors every term of (C.6) is a plane wave of energy $\sum k_j^2$ and momentum $\sum k_j$, as in [Theorem 11.3](#). ■

The two sets do different jobs. The k_j are momenta and carry the energy (C.18). The λ_α are dimensionless, as many as there are down spins, and carry *none* of it — they enter (C.16) only as phase shifts. That is spin–charge separation in its rawest form (a reading draws two states).

The counting. In logarithms,

$$k_j L + 2 \sum_{\alpha} \arctan(2(\xi_j - \lambda_\alpha)) = 2\pi I_j, \quad \sum_j \arctan(2(\lambda_\alpha - \xi_j)) - \sum_{\beta \neq \alpha} \arctan(\lambda_\alpha - \lambda_\beta) = \pi J_\alpha, \quad (\text{C.20})$$

with $I_j \in \mathbb{Z} + \frac{M}{2}$ and $J_\alpha \in \mathbb{Z} + \frac{N-M+1}{2}$, and $K = (2\pi/L)(\sum_j I_j + \sum_\alpha J_\alpha)$. **The parities are part of the equations:** a wrong one solves (C.16) with an extra minus sign, invisible in (C.20) · *checked*.

Example C.7 ($N = 2$: the two channels, again). Take $N = 2$, $M = 1$. (C.17) reads $(\lambda - \xi_1 + \frac{i}{2})(\lambda - \xi_2 + \frac{i}{2}) = (\lambda - \xi_1 - \frac{i}{2})(\lambda - \xi_2 - \frac{i}{2})$, whose only root is $\lambda = \frac{1}{2}(\xi_1 + \xi_2)$; then

$\xi_1 - \lambda = \frac{1}{2}(\xi_1 - \xi_2)$ and (C.16) becomes

$$e^{ik_1 L} = \frac{k_1 - k_2 + i\epsilon}{k_1 - k_2 - i\epsilon},$$

exactly Lecture 11's Lieb–Liniger equation (11.9) for two particles. The other $M = 1$ state is the descendant $\lambda \rightarrow \infty$, where the right-hand side of (C.16) tends to 1: free fermions. The two channels of Figure C.1, recovered from the nesting · *checked*.

Whether these solutions exhaust the spectrum is a separate question, and the answer is Lecture 10's reading's for the Heisenberg chain: they do not, without the complex ones. The reading at the end says what is known · *cited: Takahashi ch. 12; Essler et al. ch. 4.*

C.5 The Hubbard model

The Hubbard model's Bethe equations are *stated* here, and their derivation *cited*; the tutorial solves the lattice two-body problem, which fixes the correspondence with § C.4, sign included.

On a ring of L sites, L even, with $c_{j\sigma}^\dagger$ creating an electron of spin σ at site j and $n_{j\sigma} = c_{j\sigma}^\dagger c_{j\sigma}$,

$$H = -t \sum_{j=1}^L \sum_{\sigma} (c_{j\sigma}^\dagger c_{j+1,\sigma} + c_{j+1,\sigma}^\dagger c_{j\sigma}) + U \sum_{j=1}^L n_{j\uparrow} n_{j\downarrow}, \quad (\text{C.21})$$

and we set $t = 1$: the Yang–Gaudin gas put on a lattice, with the dispersion $-2 \cos k$, which is not quadratic, so there is no Galilean invariance.

Theorem C.8 (Lieb and Wu 1968). *The regular eigenstates of (C.21) with N electrons and M down spins — those with finite, pairwise distinct roots, and $2M \leq N \leq L$ — are given by charge momenta $k_1, \dots, k_N$ and spin rapidities $\Lambda_1, \dots, \Lambda_M$ solving*

$$e^{ik_j L} = \prod_{\alpha=1}^M \frac{\sin k_j - \Lambda_\alpha + iU/4}{\sin k_j - \Lambda_\alpha - iU/4}, \quad (\text{C.22})$$

$$\prod_{j=1}^N \frac{\Lambda_\alpha - \sin k_j + iU/4}{\Lambda_\alpha - \sin k_j - iU/4} = \prod_{\beta \neq \alpha}^M \frac{\Lambda_\alpha - \Lambda_\beta + iU/2}{\Lambda_\alpha - \Lambda_\beta - iU/2}, \quad (\text{C.23})$$

with $E = -2 \sum_j \cos k_j$ · *cited: Lieb and Wu (1968); Essler et al. ch. 3. They are not all the states, and not all the rest are symmetry descendants either (the reading gives one of each on a six-site ring).*

With $\Lambda_\alpha = c\lambda_\alpha$ they are (C.16)–(C.17) under

$$k_j \mapsto \sin k_j, \quad c \mapsto \frac{U}{2t}, \quad (\text{C.24})$$

with *that* sign: with $c \mapsto -U/2t$ none of the eighteen singlet states of the two-electron $L = 6$ ring satisfies (C.22); with it, all do · *checked*. Shastry's R -matrix behind them is not of difference form, which Lecture 14 and Lecture 15 used (the reading). For κ species the construction nests $\kappa - 1$ times, one level per node of the Dynkin diagram $A_{\kappa-1}$ (the reading).

Elastic non-diffractive scattering: the asymptotic momenta are only permuted, and a collision leaves a displacement ([Theorem 12.2](#), cited) $\rightarrow$ the momenta are still only permuted (([C.18](#))) and a collision still leaves a phase, but the labels are *transported* by X ([Figure C.2](#)), so the ring condition is an eigenvalue problem and not a number (*type A; the Part II side proved, the Part I side cited*)

C.6 What it buys

- *Nesting*: a matrix S-matrix on a ring is an eigenvalue problem, and the eigenvalue problem is an inhomogeneous Heisenberg chain whose sites are the particles ([Propositions C.4](#) and [C.5](#)), so the algebraic Bethe ansatz solves it.
- *Two families of rapidities*: the momenta carry all of the energy, the spin rapidities none of it ([Theorem C.6](#)).
- *The Hubbard model's equations*, the same system under $k \mapsto \sin k$, $c \mapsto U/2t$ ([Theorem C.8](#), cited).

C.7 Caveats, in one place

- *Status*. [Theorem C.2](#) is proved, except that its reduction of the $N!$ conditions to N on one vector is sketched here and proved in a reading; [Propositions C.4](#) and [C.5](#) are proved; [Theorem C.6](#) is proved given that reduction and the algebraic Bethe ansatz of [Lecture 15](#) with unequal inhomogeneities, checked here at $N = 4$. The Lieb–Wu equations are cited; their $N = 2$ case is derived in the tutorial.
- *Completeness is not claimed*, for the gas or for the lattice: the nested equations need complex spin roots, and the regular Lieb–Wu solutions miss states (the readings).
- *The fermionic sign and the Hubbard sign are not conventions*; each wrong choice gives wrong numbers.

Reading

C.8 The auxiliary chain drawn; which shift; two states drawn

Remark C.9 (Which shift, and why it is not free). Write the spin rapidities as $\lambda_\alpha = v_\alpha + \frac{i}{2}$. [Lecture 15](#)'s homogeneous chain names the same roots $v_\alpha = -\lambda_\alpha - \frac{i}{2}$ (([15.6](#))), so *this lecture's λ_α is minus [Lecture 10](#)'s*. There the sign was a labelling: the homogeneous system is invariant under $\lambda \rightarrow -\lambda$ and the two choices name a mirror doublet, which only the *vector* tells apart. Here the ambiguity generically does not arise — for generic inhomogeneities the spin equations are not invariant and the mirrored roots do not solve them, although a symmetric set such as $\xi = (-1, 0, 1)$ is invariant (at $M = 1$ both roots $\pm\sqrt{15}/6$ solve it) — but the check script builds $B(v_1) \cdots B(v_M) |0\rangle$ anyway and confirms it is the eigenvector used · *checked*.

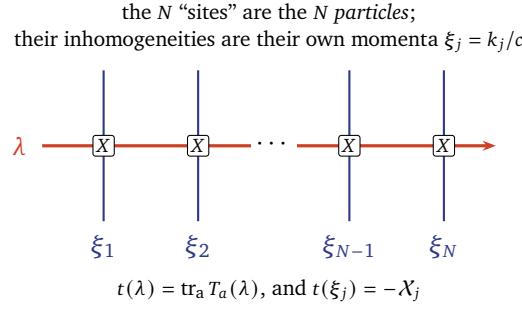


Figure C.3: The auxiliary chain of Definition C.3, drawn as Lecture 14 drew the Heisenberg chain — with the one difference that is the commonest place to stall: its sites are not lattice sites but the particles of a gas on a continuous ring, each at its own rapidity. Setting λ equal to one of them collapses the horizontal line onto a vertical one (Proposition C.4), which is why the commuting family contains the N periodicity operators.

Reading

C.9 The Hubbard model's symmetry, and a check

The equations of Theorem C.8 are checked against exact diagonalisation of a six-site ring, where 20 of the 21 states of the singlet sector are regular Bethe states and the twenty-first is the η -pair of Remark C.10. Eighteen of the twenty — those with $K \neq \pi$, complex roots included — are checked to 10^{-14} by reconstructing the roots from (K, E) ; that reconstruction degenerates at $K = \pi$, so the two states there are checked directly against the equations instead · *checked*. The $SO(4)$ commutators below are computed exactly, on one even ring; they are why L is even.

Remark C.10 ($SO(4)$). (C.21) has more symmetry than the spin $su(2)$. Let $\eta^\dagger = \sum_j (-1)^j c_{j\uparrow}^\dagger c_{j\downarrow}^\dagger$ create a pair on a site, with the staggering. Then

$$[H, \eta^\dagger] = U\eta^\dagger, \quad [H, S^+] = 0, \quad [\eta^\dagger, S^+] = 0,$$

all three for **even** L : the hopping terms cancel in pairs along the ring, and the closing bond cancels only if $(-1)^L = 1$. At $L = 3$, $U = 7$ the commutator misses $U\eta^\dagger$ by 2 · *checked*. So $H - \frac{U}{2}\hat{N}$ carries a second $su(2)$ commuting with the first. As Lie algebras $su(2) \oplus su(2) \cong so(4)$; the $\mathbb{Z}_2$ of the familiar statement $(SU(2) \times SU(2))/\mathbb{Z}_2 \cong SO(4)$ lives at the group level, and no commutator can see it.

For the printed H the η 's are spectrum-generating, not symmetries: $[H, \eta^\dagger] = U\eta^\dagger$ moves the energy rather than fixing it, and it is $H - \frac{U}{2}\hat{N}$ whose degeneracies $SO(4)$ organises · *cited*: Yang (1989); Essler et al. ch. 2. The staggering makes $\eta^\dagger$ anticommute with the translation, so it raises the momentum by π , and $\eta^\dagger |0\rangle$ is an exact eigenstate of (C.21) of energy U at $K = \pi$ · *checked* — and outside Theorem C.8, because at $K = \pi$ every finite Bethe pair has $\cos k_1 + \cos k_2 = 0$, so $E = 0$, while exact diagonalisation at $L = 6$ returns 0, 0 and U .

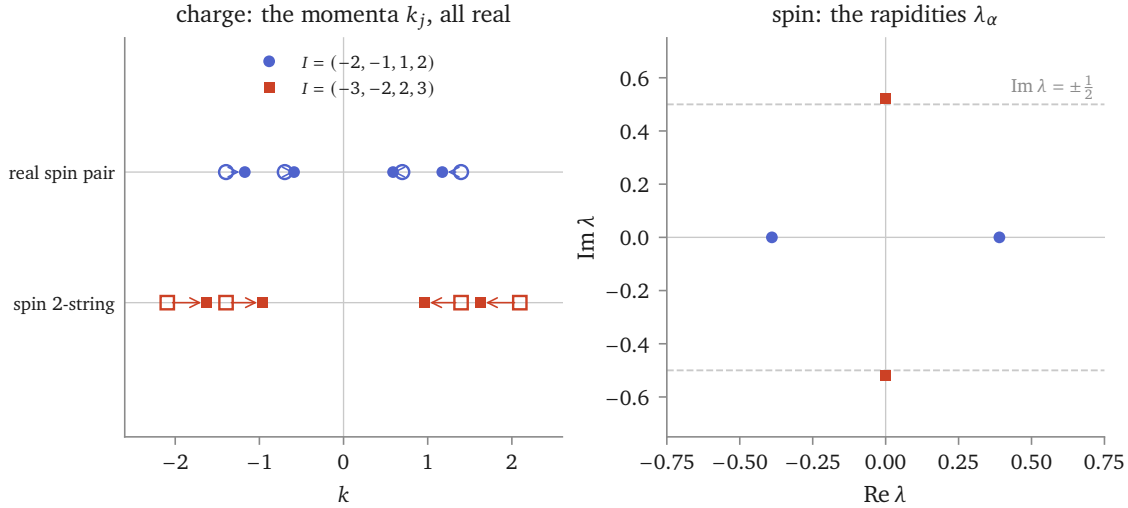


Figure C.4: Two spin-singlet Bethe states at $N = 4$, $M = 2$, $L = 9$, $c = 3$, total momentum zero. Left: both have four *real* momenta, so this panel is one axis and the two rows are only a device for keeping the states apart. Each is drawn against its own $2\pi I_j/L$ (open marks, arrow the shift), and the shift is inward for both. **The two states do not share their charge quantum numbers:** (C.20) gives $(-2, -1, 1, 2)$ for one and $(-3, -2, 2, 3)$ for the other, which no residual of (C.16) can see. Right: one state has a real spin pair, the other a 2-string on the imaginary axis — a bound state of two spin waves — with energy 7.17 against 3.44. From `scripts/figures/nested_roots.py`.

Reading

C.10 Higher rank: where the Dynkin diagram comes in

Nothing in § C.3 used that the label space is $\mathbb{C}^2$. For κ species the two-body matrix is still $X(u) = R(-u)/(u + i)$ with $R(u) = u - iP$ on $\mathbb{C}^\kappa \otimes \mathbb{C}^\kappa$, it still obeys the Yang–Baxter equation · *checked*, and Theorem C.2 and Proposition C.4 go through word for word. What changes is § C.4. The auxiliary *monodromy* is now $\kappa \times \kappa$ in the auxiliary index — the transfer matrix $t = \text{tr}_a T_a$ still acts on the κ^N -dimensional label space — its reference state breaks sl_κ to $sl_{\kappa-1}$, and its first row carries $\kappa - 1$ creation operators $B_2(\lambda), \dots, B_\kappa(\lambda)$ in place of Lecture 15’s single B . A Bethe vector $B_{i_1}(\lambda_1) \cdots B_{i_M}(\lambda_M) |0\rangle$ therefore carries a vector of indices, and choosing that vector is itself a Bethe-ansatz problem of rank $\kappa - 1$, whose auxiliary problem needs another, and so on — $\kappa - 1$ levels, the last of them scalar. That is why the procedure is called nesting, and the extra creation operators are what Lecture 15’s reading pointed at.

The bookkeeping is the Lie algebra’s. $R(u) = u - iP$ commutes with $A \otimes 1 + 1 \otimes A$ for every $A \in sl_\kappa$ · *checked*, so the hierarchy is sl_κ -invariant, and the $\kappa - 1$ levels are the nodes of the Dynkin diagram $A_{\kappa-1}$: level a carries M_a rapidities $\lambda^{(a)}$, and the equations couple level a to level b through the Cartan matrix,

$$\prod_{j=1}^N \frac{\lambda_\alpha^{(a)} - \xi_j + \frac{i}{2}\delta_{a1}}{\lambda_\alpha^{(a)} - \xi_j - \frac{i}{2}\delta_{a1}} = \prod_{b=1}^{\kappa-1} \prod_{(b,\beta) \neq (a,\alpha)} \frac{\lambda_\alpha^{(a)} - \lambda_\beta^{(b)} + iC_{ab}/2}{\lambda_\alpha^{(a)} - \lambda_\beta^{(b)} - iC_{ab}/2}, \quad C = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \text{ for } A_2, \quad (\text{C.25})$$

the source appearing only at the node the particles attach to — the left-hand side is 1 at every other node, $\delta_{a1} = 0$ — and the term $(b, \beta) = (a, \alpha)$ **excluded**: with $C_{aa} = 2$ it would

contribute $i/(-i) = -1$ and turn the $\kappa = 2$ case into $1 = -1$ · *checked*. Equation (C.17) is that $\kappa = 2$ case (Figure C.5); the A_2 pair, whose second equation contains no ξ at all, is in the reading · *cited: Sutherland (1968), (16)–(17); Babelon, de Vega and Viallet (1982)*.

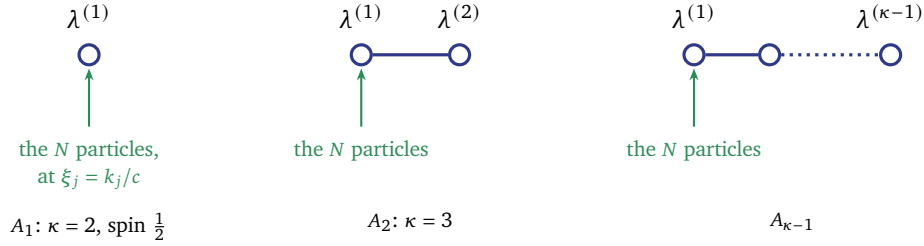


Figure C.5: The levels of the nesting are the nodes of the Dynkin diagram and the gas hangs off the first. Each node carries its own rapidities $\lambda^{(a)}$; nodes joined by an edge have $C_{ab} = -1$ and appear in each other's equations, unjoined ones have $C_{ab} = 0$ and do not. For $\kappa = 2$ there is one node, (C.17). Only that first node sees the ξ_j .

The same diagram indexes the other series once the label space carries a different representation, and then the representation data — weights, matrix elements of the generators — is what the computation needs; the rep-engine package supplies it exactly, and the check script takes sl_3 's fundamental from it.

In the reading of optional lecture B the root systems A_{N-1} , B_N , C_N , D_N index *classical* Calogero–Moser models, one interaction per positive root; here $A_{\kappa-1}$ indexes the levels of a *quantum* Bethe ansatz, one set of rapidities per node. The same combinatorial object, two unrelated uses, and no theorem in this course connecting them.

Dictionary

The root system of sl_N , indexing the classical Calogero–Moser models — one pair interaction per positive root (optional lecture B, reading) $\rightarrow$ the Dynkin diagram of $A_{\kappa-1}$, indexing the levels of the nested Bethe ansatz for κ -component particles — one set of rapidities per node, coupled by the Cartan matrix (C.25) (*type A, cited*)

Reading

C.11 Shastry's R-matrix, and two transcription traps

The R-matrix. Shastry (1986) built a two-dimensional vertex model whose transfer matrix commutes with (C.21), by taking two six-vertex models at their free-fermion point and coupling them with a *local* diagonal factor $\exp(hC^\uparrow C^\downarrow/2)$ whose parameter is tied to the spectral one by $\sinh 2h = u \sin 2z$. Two things are worth separating. Jordan–Wigner is the transformation between the fermions and the spin variables, not the coupling. And the loss of difference form is *not* what stops t at a special point from being the shift operator: that is *regularity*, $R(z, z) \propto P$, which Shastry's matrix has. What difference form buys is one R generating a whole commuting family from a single argument, which is what [Lecture 14](#)'s construction used · *cited: Shastry (1986); Essler et al. ch. 12*.

The other nodes. For A_2 the second node of (C.25) reads

$$1 = \prod_{\beta \neq \alpha} \frac{\lambda_\alpha^{(2)} - \lambda_\beta^{(2)} + i}{\lambda_\alpha^{(2)} - \lambda_\beta^{(2)} - i} \prod_\gamma \frac{\lambda_\alpha^{(2)} - \lambda_\gamma^{(1)} - \frac{i}{2}}{\lambda_\alpha^{(2)} - \lambda_\gamma^{(1)} + \frac{i}{2}},$$

with no ξ in it at all: the second level sees the particles only through the first. Sutherland's original writes both levels with an external minus and the self-root included, which is the same equation once the self-factor -1 is taken across · *cited: Sutherland (1968), (16)–(17)*.

The two statistics. The bosonic $X_B = R(u)/(u+i)$ of Lecture 13 also has $X_B(0) = -P$, because $R(0) = -iP$: the two statistics are told apart by $u \rightarrow -u$, not by the value at the origin, and Proposition C.4 holds for both. What they do not share is the periodicity condition · *checked*.

An exceptional state. Theorem C.8 is restricted to regular solutions, and its escape clause is not decorative. At $L = 6$, $N = 2$, $U = 2\sqrt{3}/3$ the symmetric wavefunction $e^{ik_0(x+y)}(3 - |x - y|)$ with $k_0 = 7\pi/6$ is an exact eigenstate of energy $2\sqrt{3}$ at $K = \pi/3$. Any two-root representation of it must satisfy $E = -4 \cos(K/2) \cos(\frac{1}{2}(k_1 - k_2))$, which here forces $k_1 = k_2$: at coincident roots the cusp condition gives $A' = -A$ and the ordinary ansatz vanishes, so this state is a regularised limit of a colliding pair, reached as $q \rightarrow 0$ along $k = k_0 \pm iq$. Nor is it a descendant: η annihilates it and the spin generators preserve the total spin · *checked*.

The Hamiltonian. The reference book writes (C.21) in its particle–hole symmetric form, $U \sum (n_\uparrow - \frac{1}{2})(n_\downarrow - \frac{1}{2})$, so its energy carries an extra $\frac{U}{4}(L - 2N)$; its coupling is $u = U/4t$, ours is written out; and its spin equation is the reciprocal of (C.23). All three differences are invisible in a formula copied out of context · *checked*.

The gap. § C.12's Δ_μ is the chemical-potential discontinuity. The same book's Δ is the single-holon threshold, exactly half of it, and is printed in Bessel form with $u = U/4t$. Which gap is meant is not a detail: the two differ by a factor of two at every coupling · *checked*.

Reading

C.12 Half filling, the Mott gap, and spin–charge separation

At half filling, $N = L$, the Hubbard model is an insulator for every $U > 0$. Lieb and Wu's own paper established it from (C.22)–(C.23) in the thermodynamic limit: the *charge gap*, meaning the discontinuity $\mu_+ - \mu_-$ of the chemical potential, is

$$\Delta_\mu = \frac{16t^2}{U} \int_1^\infty \frac{\sqrt{y^2 - 1}}{\sinh(2\pi t y/U)} dy,$$

positive for every $U > 0$, tending to $U - 4t$ from above and to $(8/\pi)\sqrt{U}t e^{-2\pi t/U}$ as $U \rightarrow 0$ · *cited: Lieb and Wu (1968), (21)–(23)*. Say which gap. The reference book's Δ (its (7.18)) is the *single-holon* threshold, exactly half of this, and is written in Bessel form with $u = U/4t$; the two agree to every digit once converted · *checked*. There is no metal–insulator transition at finite U in one dimension; the transition is at $U = 0^+$. The spin sector meanwhile stays gapless, and at large U the half-filled model reduces to the Heisenberg antiferromagnet of Lecture 10 with $J = 4t^2/U$: second-order hopping of an electron onto an occupied site and back costs U and gains t^2 , and it can only happen if the two spins are antiparallel. On a four-site ring, where the Heisenberg ground energy is $-3J = -12/U$, the two agree to 0.62% at $U = 40$ and 0.039% at $U = 160$ · *checked*.

That is spin–charge separation, and it is visible already in § C.4: the energy (C.18) does not contain λ_α . In the thermodynamic limit the two sectors acquire different sound velocities — v_c from the k 's and v_s from the λ 's — and an added electron's charge and spin propagate apart. The Luttinger-liquid description of the resulting low-energy theory, and the angle-resolved photoemission experiments on quasi-one-dimensional conductors that

look for the two dispersing features, are the obvious continuation · *cited: Essler et al. ch. 8; Giamarchi, Quantum Physics in One Dimension.*

Reading

C.13 What is not claimed about the gas

That the solutions of (C.16)–(C.17) exhaust the spectrum is *not* proved here, and the situation is the one [Lecture 10](#)’s reading (§ 10.10) described for the Heisenberg chain: real roots do not suffice. Here the complex ones live in the *spin* sector — strings of λ ’s, one of which is drawn in [Figure C.4](#) — while for $c > 0$ the momenta k_j stay real. [Proposition 11.4](#) proves that for the *spinless* gas; for the spinful one it is Yang’s and Gaudin’s statement, and it is not proved here. Completeness is a counting statement with known exceptions · *cited: Takahashi ch. 12; Essler et al. ch. 4.* The thermodynamics built on that string hypothesis — the analogue of [Lecture 17](#) for two coupled families — is a genuine subject and is not attempted.

Reading

C.14 Bosons with a label, and the attractive gas

Two variants cost one line each and are worth knowing.

Two-component bosons. (C.3) becomes $Y_B = S\Pi_+ - \Pi_-$, so $X_B(u) = R(u)/(u + i)$ — the R -matrix itself rather than $R(-u)$, and the interacting channel is now the symmetric one. The auxiliary chain is again of Heisenberg type, the fermionic $(-1)^{N-1}$ of (C.9) is absent, and the nested equations come out with the shifts reflected. Substituting the bosonic X into the fermionic condition without touching the statistics sign gives $e^{ikL} = -1$ in the singlet: antiperiodic quantisation, and a spectrum wrong by half a level · *checked.*

Attraction. For $c < 0$ the k_j need no longer be real ([Proposition 11.4](#) assumed $c > 0$), and the gas forms bound pairs of one up and one down spin — the one-dimensional analogue of a Cooper pair. In the *thermodynamic limit* the pair is the string $k = c\lambda \pm ic/2$ · *cited: Gaudin, The Bethe Wavefunction, §11.4, where the string is presented as a large-system approximation.* It is not exact on a finite ring: at $N = 2$, $K = 0$, $c = -g < 0$ the even bound wave is $\cosh q(x - L/2)$ with $2q \tanh(qL/2) = g$, so $q > g/2$; at $g = 1$, $L = 2$ the true q is 0.7717 against the string’s 0.5 · *checked.* Nothing in § C.3 used the sign of c ; everything in § C.4 about *real* solutions did.

Tutorial

C.15 Two electrons on a lattice, from scratch

The Lieb–Wu equations were stated, not derived. Here is the $N = 2$ case in full, on a lattice of L sites, and it is enough to fix (C.24) including its sign.

Take one up electron and one down electron; the spin singlet has a symmetric spatial wavefunction $\psi(x_1, x_2)$. For $x_1 \leq x_2$ put

$$f(x_1, x_2) = A e^{i(k_1 x_1 + k_2 x_2)} + A' e^{i(k_2 x_1 + k_1 x_2)},$$

set $\psi = f$ there and $\psi(x_1, x_2) = f(x_2, x_1)$ elsewhere. Away from the diagonal the lattice Schrödinger equation is free and gives $E = -2(\cos k_1 + \cos k_2)$ for any A, A' .

The diagonal. At $x_1 = x_2 = x$ the equation reads

$$-\left[\psi(x-1, x) + \psi(x+1, x) + \psi(x, x-1) + \psi(x, x+1)\right] + U\psi(x, x) = E\psi(x, x),$$

and by symmetry $\psi(x+1, x) = f(x, x+1)$ and $\psi(x, x-1) = f(x-1, x)$, while the free equation for f continued to the diagonal is the same expression with every ψ replaced by f and no U . Subtracting,

$$\left[f(x+1, x) + f(x, x-1) - f(x-1, x) - f(x, x+1)\right] + Uf(x, x) = 0.$$

Every term carries $e^{i(k_1+k_2)x}$; cancelling it leaves a linear relation whose coefficients are, with $s_j = \sin k_j$,

$$\left[2i(s_1 - s_2) + U\right]A + \left[-2i(s_1 - s_2) + U\right]A' = 0, \quad \text{so} \quad \frac{A'}{A} = \frac{s_1 - s_2 - iU/2}{s_1 - s_2 + iU/2}$$

· *checked*. That is [Lecture 11](#)'s $S(k) = (k - ic)/(k + ic)$ with $k \mapsto s_1 - s_2$ and $c \mapsto U/2$ — ([C.24](#)), and its sign, derived rather than transcribed.

The ring. Periodicity relates the two amplitudes once more, and here it is worth doing rather than quoting: the antisymmetry sign of [Theorem C.2](#) does not appear, because we are working with the *symmetric spatial* function of the singlet. Since $x_2 < x_1 + L$, $\psi(x_1+L, x_2) = f(x_2, x_1+L)$, so $f(x_2, x_1+L) = f(x_1, x_2)$; matching the coefficients of $e^{i(k_2x_1+k_1x_2)}$ and $e^{i(k_1x_1+k_2x_2)}$ gives

$$A e^{ik_2L} = A', \quad A' e^{ik_1L} = A, \quad \text{so} \quad e^{ik_1L} = \frac{A}{A'} = \frac{s_1 - s_2 + iU/2}{s_1 - s_2 - iU/2},$$

which is ([C.22](#)) at $M = 1$ once ([C.23](#)) has been used: at $M = 1$ its right-hand side is 1 and its left-hand side forces $\Lambda = \frac{1}{2}(s_1 + s_2)$, so $s_1 - \Lambda = \frac{1}{2}(s_1 - s_2)$ and the two displayed forms agree. Multiplying the two charge equations gives $e^{i(k_1+k_2)L} = 1$: the total momentum is quantised, as it must be.

The test. Diagonalise ([C.21](#)) on $L = 6$ with one up and one down electron — a 36×36 matrix — and resolve it by total momentum and by spatial symmetry. There are 21 symmetric states. Eighteen of them, all those with $K \neq \pi$, satisfy the displayed equation to 10^{-14} at $U = 2, 5$ and 12 ; some need complex k_1, k_2 with $k_1 + k_2 = K$, which is what a bound k - Λ pair looks like · *checked*. The other three sit at $K = \pi$, where the energy $-2(\cos k_1 + \cos k_2)$ vanishes for *every* finite pair with $k_1 + k_2 = \pi$ and so cannot fix $k_1 - k_2$: there the equation has to be read directly. It gives $e^{ik_1L} = -1$, because $k_2 = \pi - k_1$ makes $s_1 = s_2$ and the ratio $(iU/2)/(-iU/2)$, so k lies on the *antiperiodic* grid $(2m+1)\pi/6$ — the pairs $\{\pi/6, 5\pi/6\}$ with $\Lambda = \frac{1}{2}$ and $\{-\pi/6, 7\pi/6\}$ with $\Lambda = -\frac{1}{2}$, which are the two $E = 0$ states and are perfectly regular solutions · *checked*; only the (K, E) route to them fails. **It is not true that any real pair summing to π solves it:** $(0, \pi)$ gives $e^{ik_1L} = 1$ against -1 . The third state, at $E = U$, is the η -pair of [Remark C.10](#) and is outside [Theorem C.8](#) altogether.

The continuum check, for comparison. [Example C.7](#)'s $N = 2$ gas can be solved without any Bethe ansatz. At zero total momentum, $k_1 = -k_2 = \kappa$, the relative coordinate obeys $-2\varphi'' + 2c\delta\varphi = E\varphi$ on a circle of circumference L with φ even, so $\varphi = \cos\kappa(x - L/2)$ and the jump condition $2\varphi'(0^+) = c\varphi(0)$ gives $\tan(\kappa L/2) = c/2\kappa$ — which is the logarithm of [Example C.7](#)'s equation. At $L = 10$, $c = 3$ both give $\kappa = 0.2775645857$ · *checked*. The factor 2 in the jump is the $\hbar = 2m = 1$ units: $\partial_1^2 + \partial_2^2 = \frac{1}{2}\partial_x^2 + 2\partial_x^2$.

Worth doing by hand. Set $U \rightarrow \infty$ in the displayed equation *along the scattering branch*, where k_1, k_2 stay real and E stays bounded: the right-hand side tends to -1 , so $k_j L \in \pi + 2\pi\mathbb{Z}$ and the two electrons quantise as free spinless fermions on a ring with antiperiodic boundary conditions. The spin has not disappeared — the state is still a singlet — but it has stopped costing energy. That is the lattice face of the impenetrable limit of [Example C.7](#). The branches with $E \sim U$ — the repulsively bound pair, $k_{1,2} = \pi \pm iq$ with $q \sim \ln(U/2)$, and the η -pair — are not hard-core states and the limit does not describe them.

Summary.

- Two channels ([C.3](#)), and a crossing is $X(u) = R(-u)/(u + i)$ ([C.4](#)).
- Periodicity is a matrix condition ([Theorem C.2](#)).
- The N conditions are one commuting family ([Propositions C.4](#) and [C.5](#)): $X_a = -t(\xi_a)$ for an inhomogeneous Heisenberg chain whose sites are the particles.
- The nested Bethe equations ([Theorem C.6](#)): ([C.16](#))–([C.17](#)), with $E = \sum k_j^2$; at $N = 2$, Lieb–Liniger in the singlet and free fermions in the triplet ([Example C.7](#)).
- Hubbard: the Lieb–Wu equations are the same system under $k \mapsto \sin k$, $c \mapsto U/2t$ ([Theorem C.8](#), cited; its sign derived at $N = 2$ in the tutorial).

References

C. N. Yang, Phys. Rev. Lett. 19 (1967) 1312; M. Gaudin, Phys. Lett. A 24 (1967) 55; E. H. Lieb and F. Y. Wu, Phys. Rev. Lett. 20 (1968) 1445; Essler, Frahm, Göhmann, Klümper and Korepin, *The One-Dimensional Hubbard Model* (CUP 2005); B. S. Shastry, Phys. Rev. Lett. 56 (1986) 1529. The readings name their other sources.